

DNA Explorer

Not Categorized
PM19

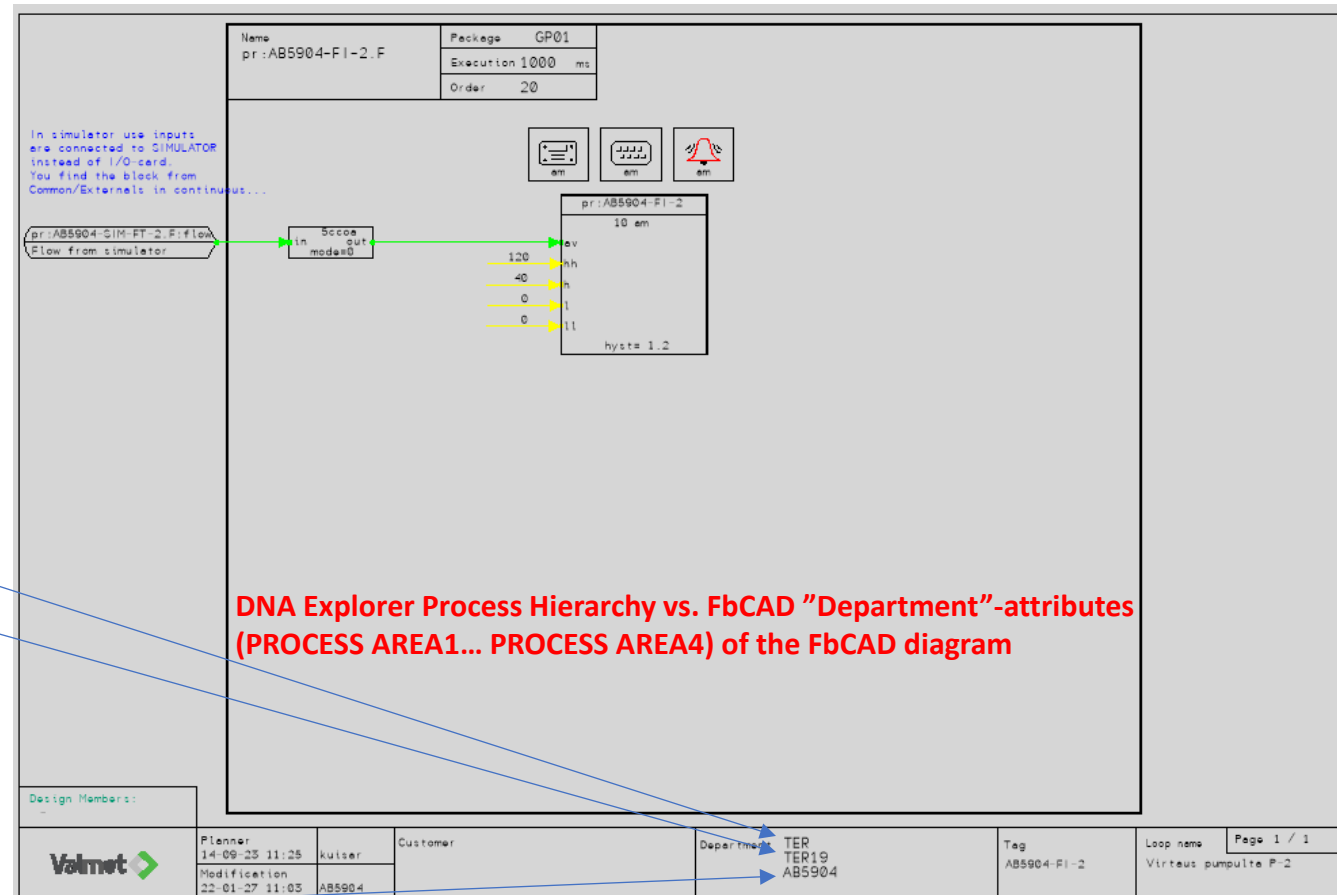
Identifier / Name Description

DNA Explorer

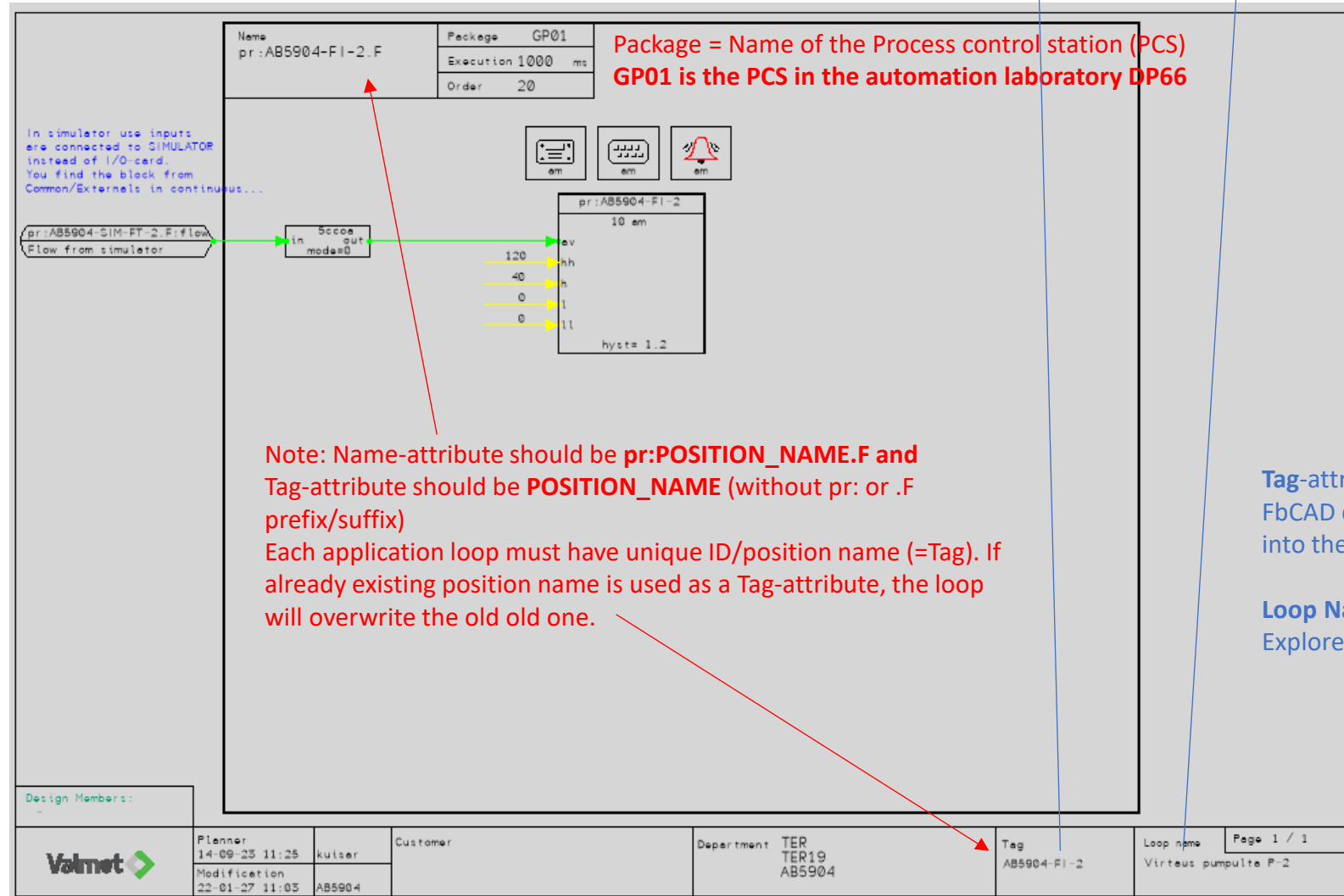
Process Area Hierarchy

- Cosme
- Design Accessories
- Diagnostic modules G1
- DP66-Vesiprosessi
- Faa
- G1
- HKatSite CtrlRoom A1
- JAMK
- localhost
- m2068
- Networks Demo
- Not Categorized
- PM19
- PM2
- SMG Simulator
- System Modules
- TER
 - TER16
 - TER17
 - TER18
 - TER19
 - AA2319
 - AA2320
 - AA2945
 - AA3003
 - AA3141
 - AA3815
 - AA3930
 - AA3936
 - AA3997
 - AA4114
 - AA4334
 - AA4449
 - AA9187
 - AA9567
 - AB5904
 - FLYTE-Teppe Flyktman

Censored

DNA Explorer Process Hierarchy vs. FbCAD "Department"-attributes
(PROCESS AREA1... PROCESS AREA4) of the FbCAD diagramMake sure, that you have chosen the right Data
Storage / Workspace (e.g TSA21KMA, teacher
will tell you the right workspace)

+	Networks Demo		Identifier	Name	Description	Check State	Io Che
-	Not Categorized		+	AB5904-FI-2	Virtaus pumpulta P-2	Externally C...	Not Ap
+	PM19		+	AB5904-FIC-1	Virtauspäättö pumpulta P-2	Externally C...	Not Ap
+	PM2		+	AB5904-HS-1	Sulkuventtiili	Externally C...	Not Ap
+	SMG Simulator		+	AB5904-HS-2	Sulkuventtiili	Externally C...	Not Ap
+	System Modules		+	AB5904-HS-3	Sulkuventtiili	Externally C...	Not Ap
-	TFR		+	AB5904-HS-4	Sulkuventtiili	Externally C...	Not Ap



Tag-attribute is the unique identifier for saving the FbCAD diagram into the DNAexplorer (or actually into the engineering database).

Loop Name -attribute is shown in the DNA Explorer (as a comment name).

INPUTS AND OUTPUTS

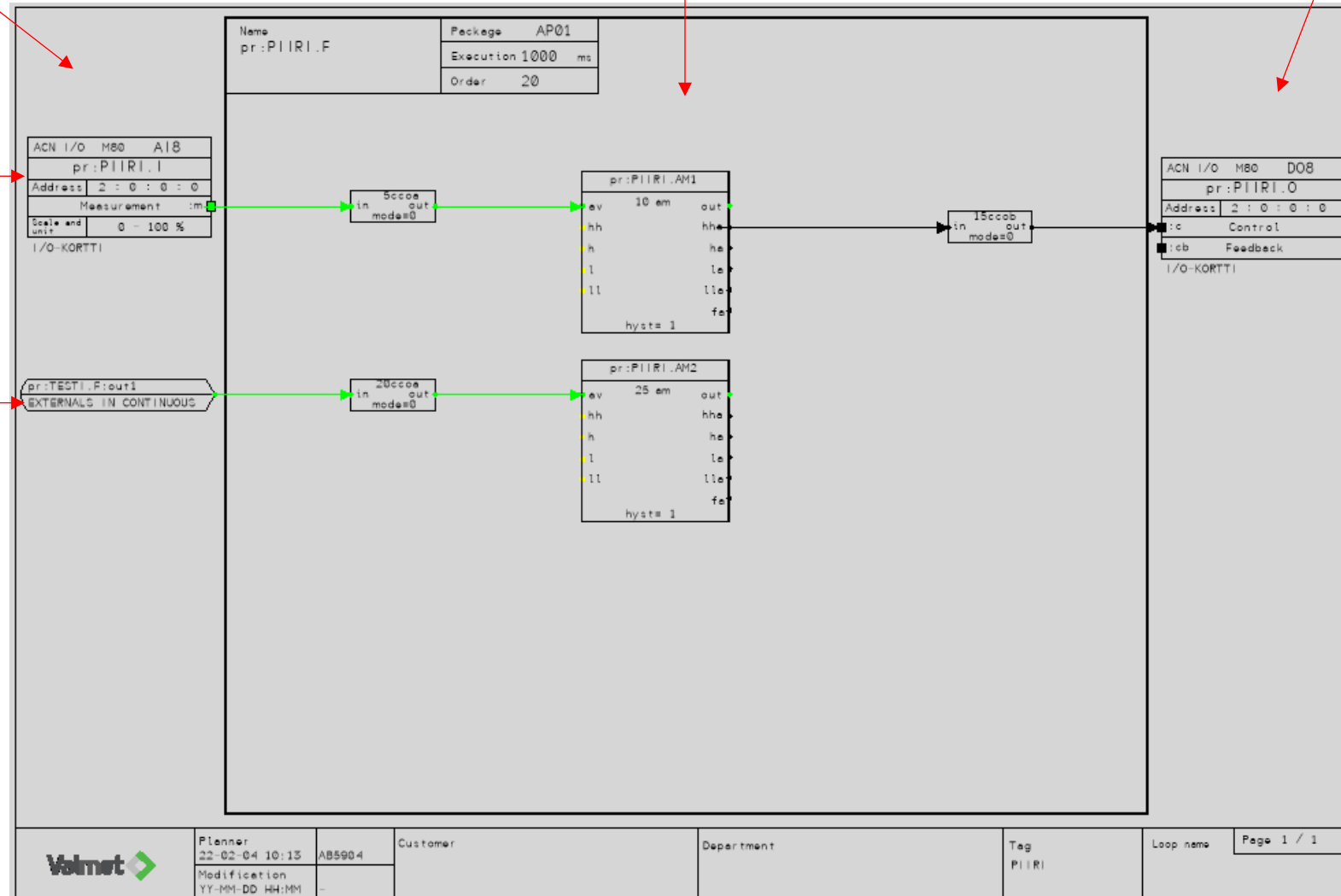
Field for external inputs

Connection field for function blocks

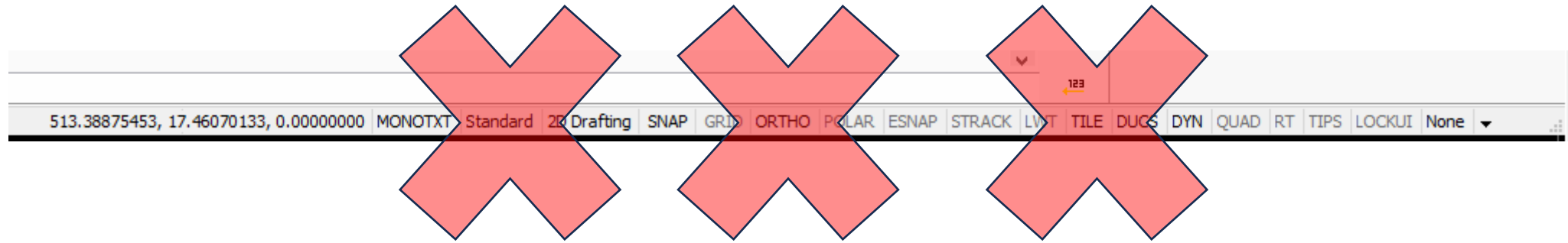
Field for external outputs

Signals from physical I/O
(or fieldbusses) is read
by using I/O-modules

Signal/variables can be
read from other
application loops by
using Externals in
continuous -blocks



Some of the rules at Valmet DNA



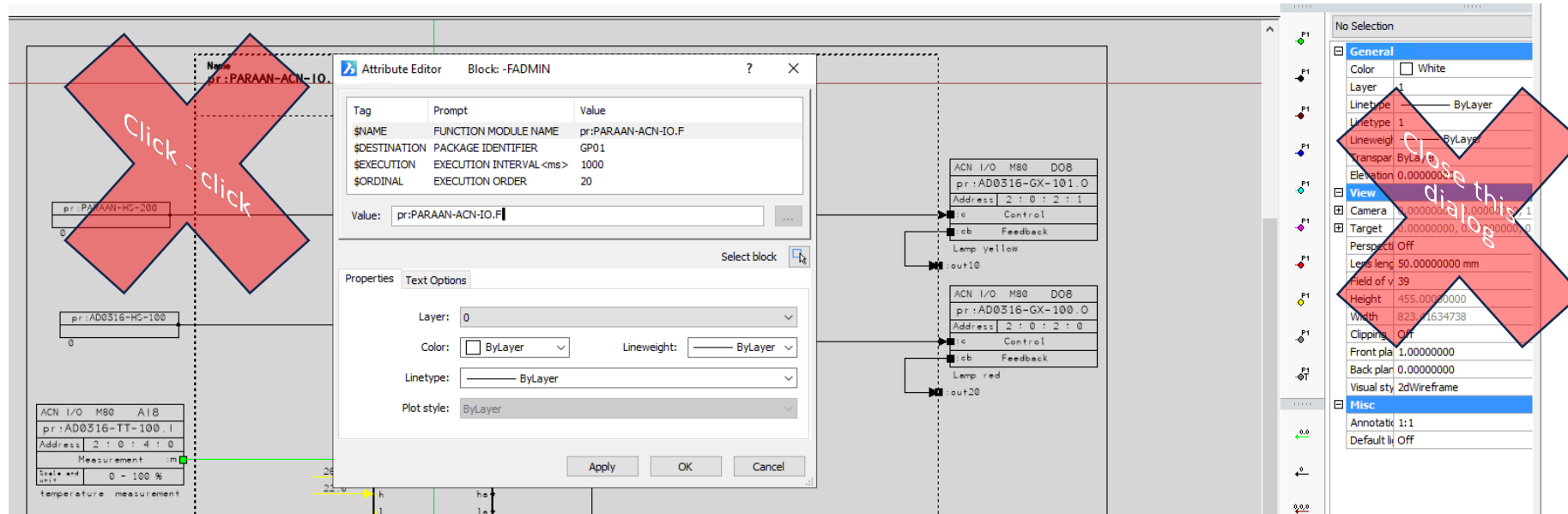
DO NOT TOUCH!
Never ever...

Some of the rules at Valmet DNA

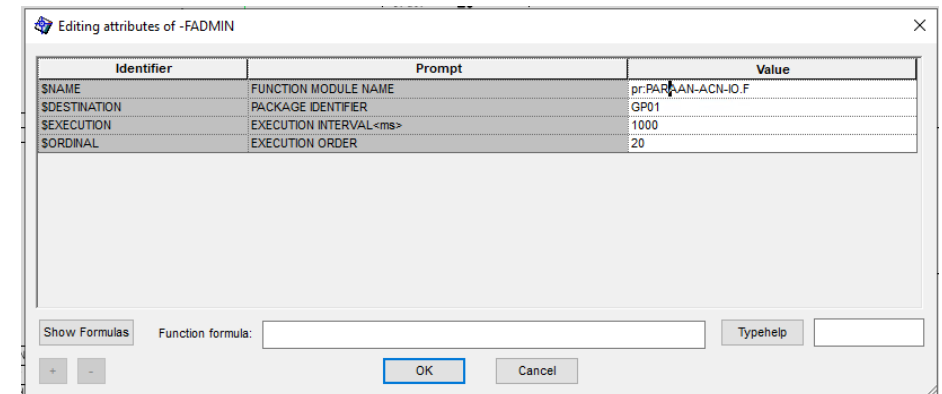
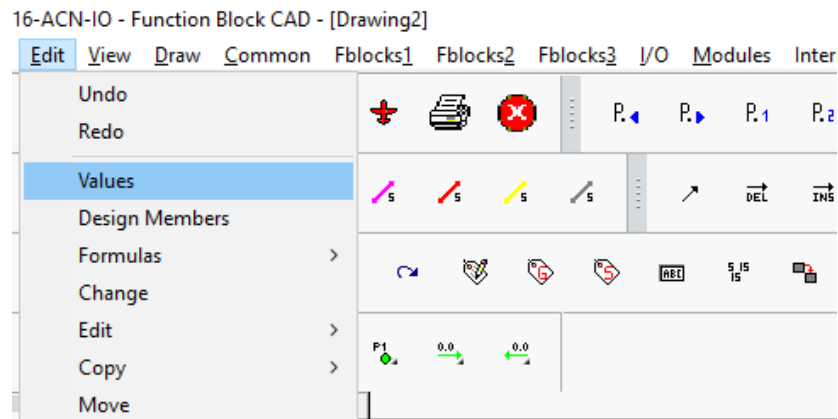
How to edit symbol values....

Never double-click and edit values in the AutoCAD dialog box!

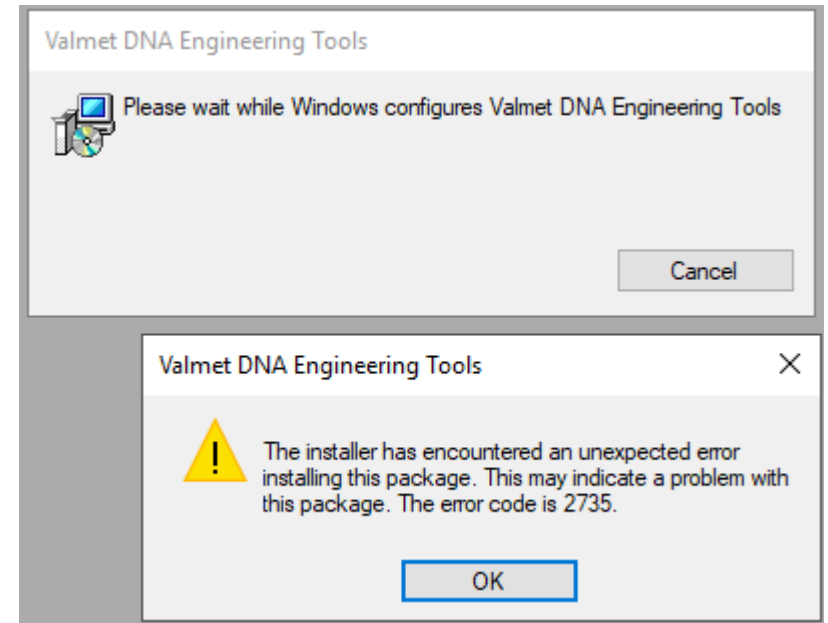
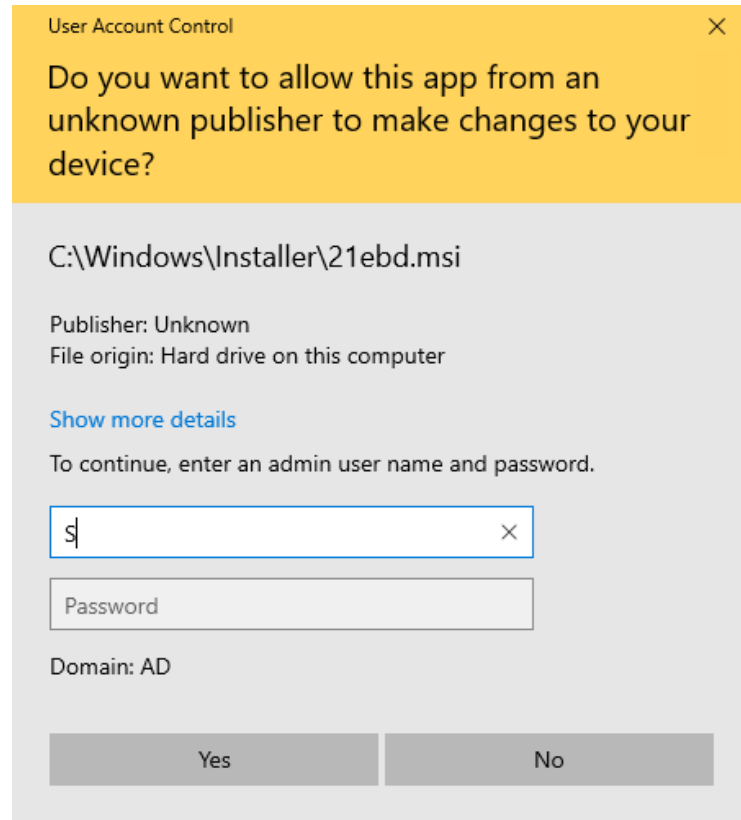
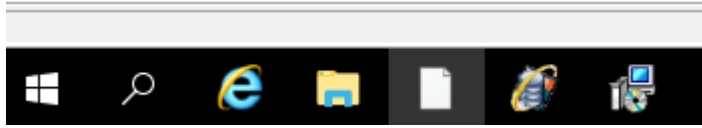
Never ever...



Always use Edit -> Values, or the pencil icon.



One annoying bug in Valmet DNA at the moment...



This also prevents other users from using the tool!!!

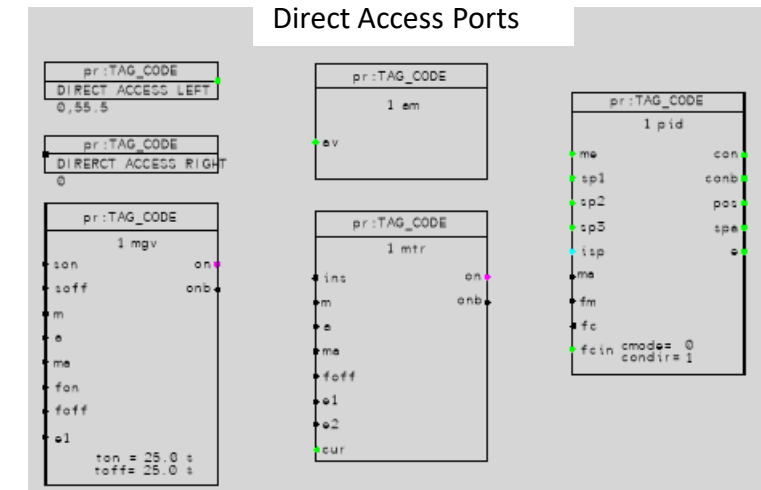
Direct Access ports and Interfaces out

“Direct Access port” and “Interfaces out” could be considered as a connection point or variable. They won’t automatically write/push data anywhere but can be used as a connection point for other FbCAD diagrams for reading the data.

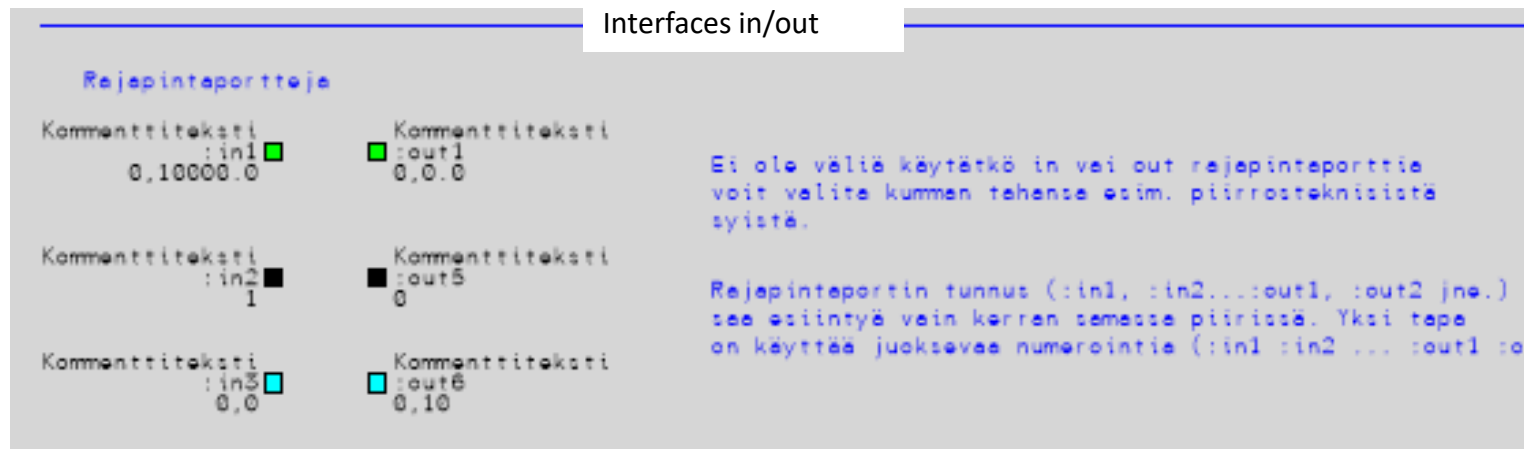
Direct Access Port must always have a **unique** identifier/position-name (pr:TAG_CODE) since it’s a global automation module known in the entire ValmetDNA automation network (=automation system). Members of a Direct Access Port can be connected to a Control Room Application by using position, operation and event modules (Control room modules - square-shaped “stamps” above the function blocks).

Some Direct Access Ports are meant for certain purposes. E.g. am, mgv, mtr and pid function blocks include direct access port but also functionality for process control.

Interfaces out can be used as a data point/port from which other application loops can read values (initial values, results of the calculations, etc.). Interfaces out is not meant to be used for transferring data to a control room application.



Name of the Direct Access Port must be unique in the whole automation system (global identifier)!!!

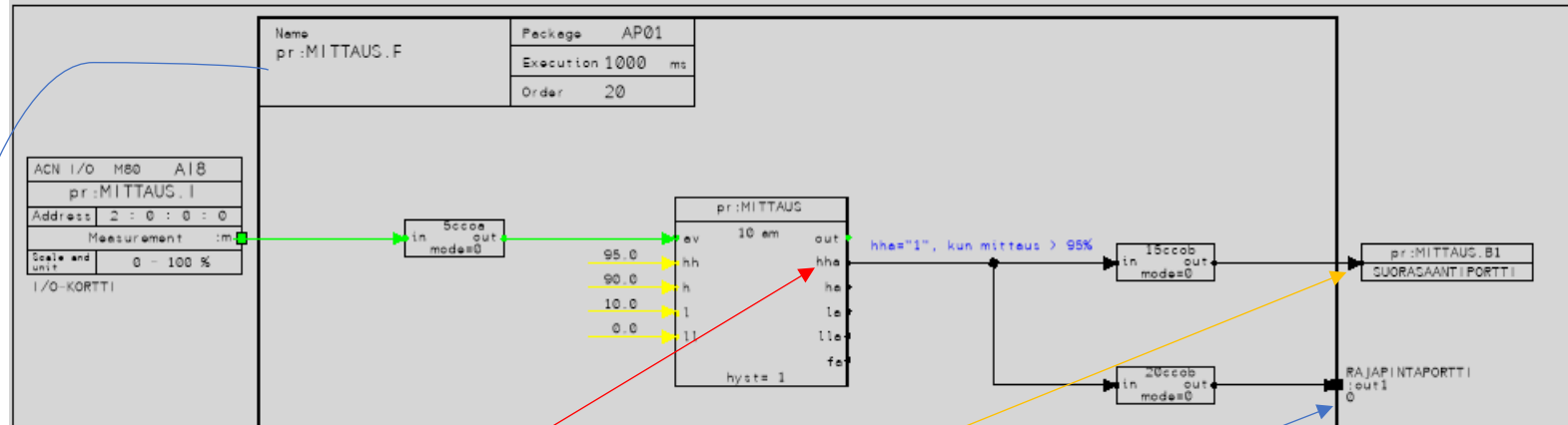


Name of the “interfaces in” or “interfaces out” can be used only once inside a single FbCAD diagram (:in1 , :in2 , :in3 , :out1 , :result1, :result2, etc).

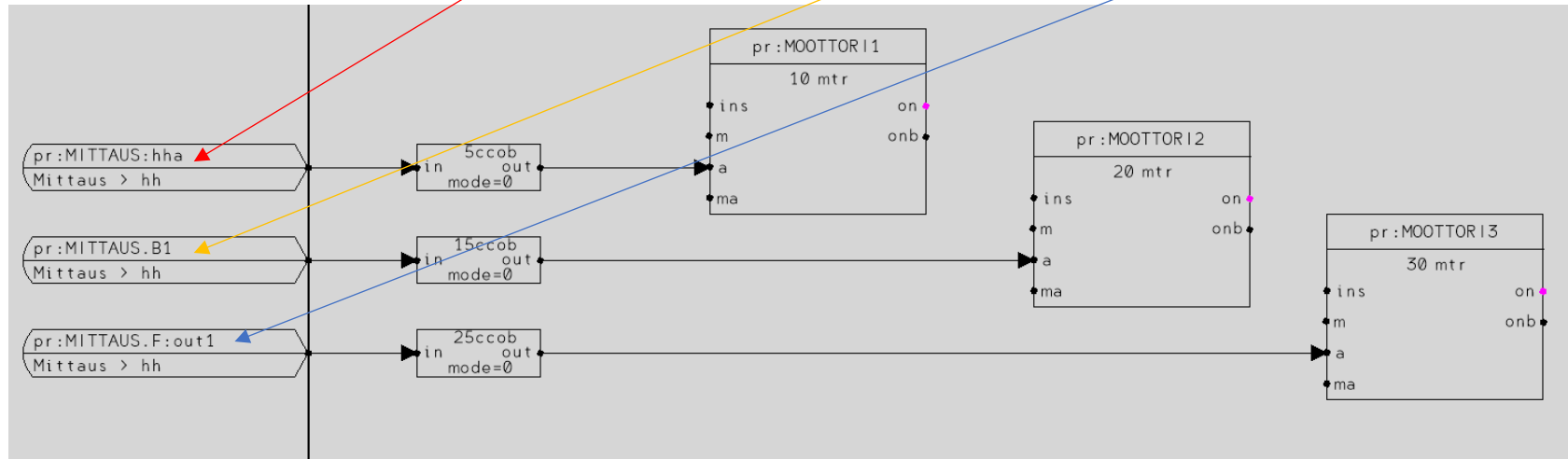
Still, it is not a global variable, but a member of an function module, so different FbCAD diagrams may include Interfaces in/out with same names because each FbCAD diagram contains function module with unique name (Name-attribute of the FbCAD diagram).

Data transfer and referencing principles between the FbCAD diagrams

Example below represents how :hha-signal (Higher High limit Alarm) could be read from one FbCAD diagram to other FbCAD diagrams. The analog measurement function block (am) has a built-in ability to perform comparison between the process value (:av) and alarm limits.



HH-alarm can be read directly from the "hha"-member (connection point) of the am-function block. Also two other ways to read are represented.



In this example all 3 MOOTTORIs would start in automatic mode when measurement reaches 95% (hh-limit of am-block)

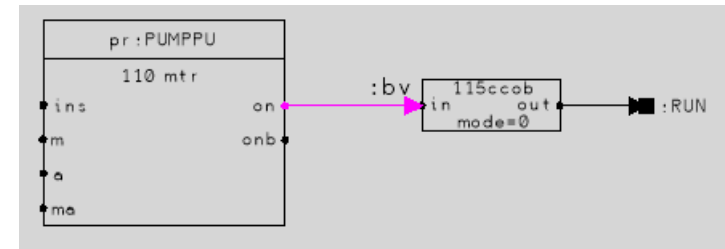
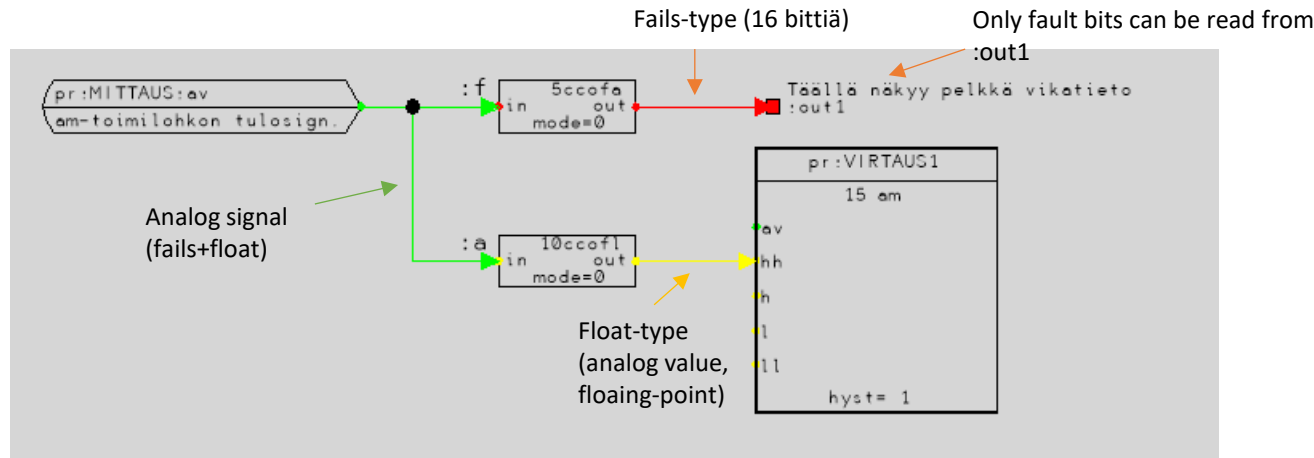
Connection points of a Direct Access Port can be read directly by using the position name and the name of the member (connection point) separated with colon: **pr:MITTAUS:hha**

If a Direct Access port includes only one variable (no separated "connectors"), it can be referred directly with its position name: **pr:MITTAUS.B1**

When referring to an Interfaces out also the name of the automation module must be included (module Name + name of the interfaces out separated by colon): **pr:MITTAUS.F:out1**

Signal manipulation and data types

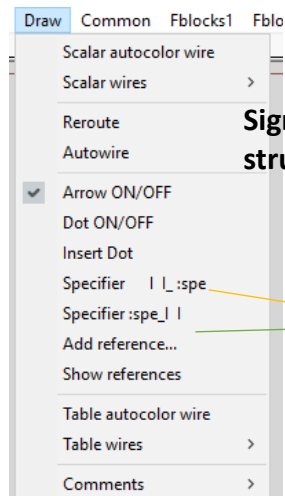
Check the Moodle –workspace for list of the data types (3. Basic loops for the water process / Attachments / Signal types.pdf). Document includes primitive types and structured types. E.g. analog signal (green) is structured signal consisting fault bits (fails – red signal) and analog value (float – yellow signal). Sometimes there is a need to separate one specific signal/information from a structured signal. Examples:



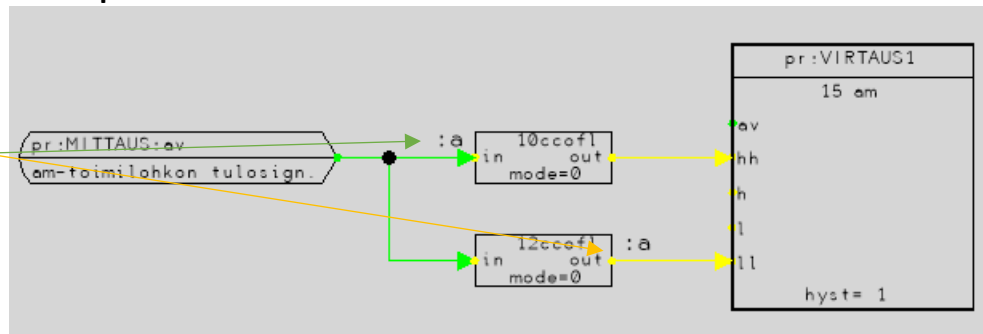
Output of the mtr-block is a structured type **bo**, which consists of:

- State information **:bv** (0/1, binary value) and
- Pulse length of the output **:pw**

For controlling an ordinary pump/motor, there is need to separate the **:bv** –signal from the structured **bo**–type

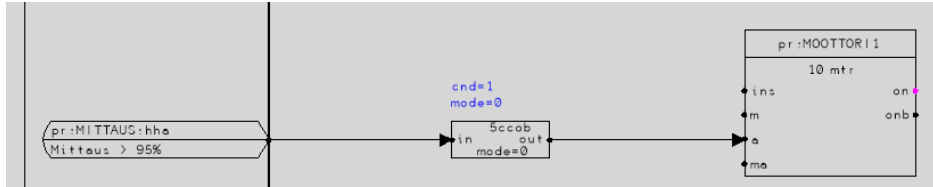


Signals can be separated from the structured types with Specifier-command.

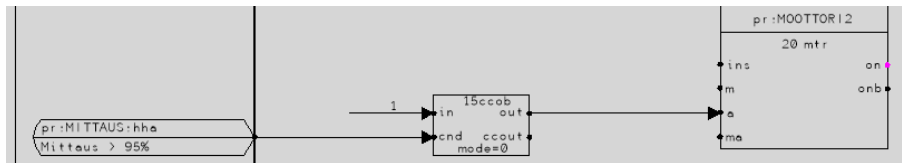


It doesn't matter on which side of the ccoX-block the separation is done. More important is to know the type of the signal after separation.

Conditional Copy – ccoX (X=data type)



Above: ccob copies :in -signal directly to :out → Motor will start in automatic mode when measurement reaches 95% and stops when measurement decreases below 95%. Why? Because parameters of the ccob has been defined so that copying condition (cnd) is always "1". When cnd="1" and Mode="0" :in will be copied to :out in every program cycle.



Above: :hha member of the measurement (am-block in an other loop) is connected cnd (copying condition), :in has constant value "1" and mode is set to "0".

Operation: when measurement reaches 95%, :in will be copied to :out → :a-member of the mtr-block has value "1". :in will be copied to :out as long as cnd="1". When measurement drops below 95%, cnd has value "0" and :in is no longer copied to :out.

ccoX-block can be configured to behave in various ways by using **cnd** and **mode** parameters.
E.g. ccob can produce a pulse that lasts only one program cycle (= edge detection, modes 1&2).
ccoX-blocks are also very commonly used to filter fault bits and for debugging purposes.

These function blocks permit conditional copying of data. In addition, the desired fault bits can be masked off or the entire fault bit part can be copied as it is.

If the copying condition is true, the block reads the in input and copies it to the out output as it is. If the copying condition is false, the output is maintained at its previous value; in other words, the block does not write anything to its output.

mode

Type: int16
Default: 0
Description: Copying mode

- 0 = copy when cnd = 1
- 1 = copy on rising edge of cnd input
- 2 = copy on falling edge of cnd input
- 3 = copy when change is detected in cnd
- 4 = copy when cnd = 0
- 5 = copy when a change from the previous copying is detected in the value part (does not apply to ccofa) or fault bit part of the in input.

NOTE!

Mode values 0 and 4 (also 5) should be used with care in connection with external copying with conditional output, because the network may become jammed if copying under the control of the function block is continued for long periods. This applies to copying between modules that have different execution cycles or are located at different Process Control Servers.

fmask

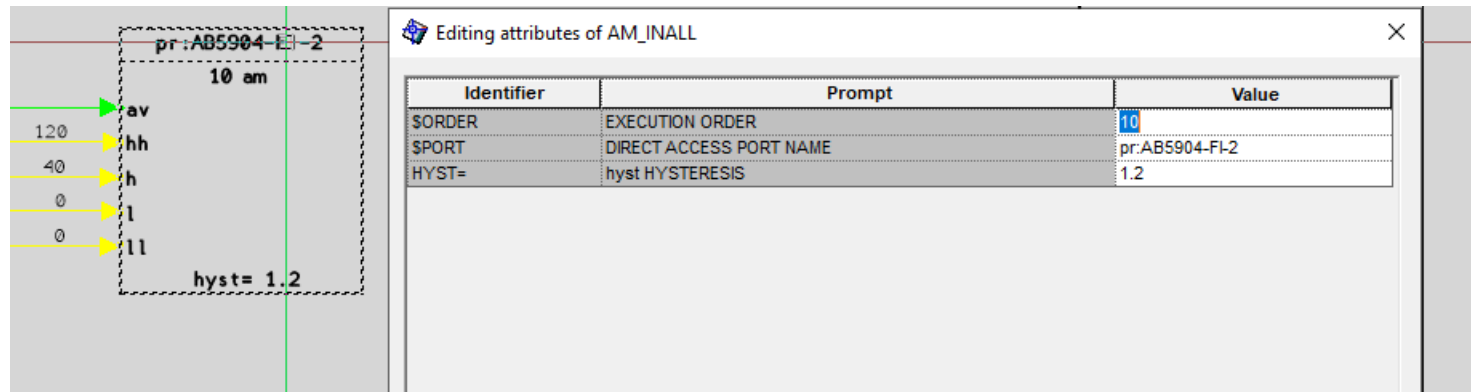
Type: uns16
Default: 0
Description: Fault bit mask

This parameter allows the masking of the input fault bits as follows:

- 0 = no masking
- 2 = mask bit ext
- 4 = mask bit ovf
- 8 = mask bit dis
- 16 = mask bit inv
- 32 = mask bit old
- 64 = mask bit der
- 128 = mask bit sex

Several fault bits can be masked by summing the corresponding values.

Analog measurement block - am



Am-block performs comparison between measured value and alarm limits and sets the corresponding alarms

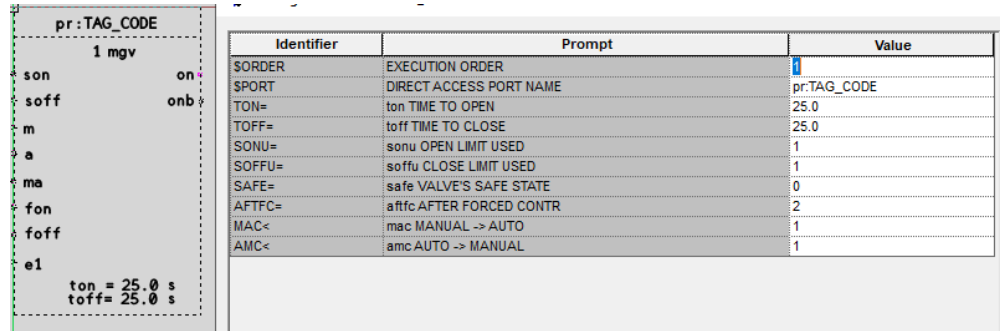
Alarm limits can be defined for:

- Higher High limit
- High limit
- Low limit
- Lower Low limit

Alarm is activated instantly when measured value crosses alarm limit. Alarm is deactivated when measured value has returned back to the allowed side of the alarm limit and difference to alarm limit is at least the amount of hysteresis.

Hysteresis is often 1% of the measurement range.

Magnetic Valve Control Function Block - mgv



The mgv function block can be used to control actuators which typically have the states OPEN and CLOSED. The basic application is to drive an auxiliary solenoid valve which drives pneumatically the main valve.

The mgv function block cannot be used to control a so-called "intermediate-position" valve (a valve that can be stopped at a position between the limits). A separate function block is provided for that purpose.

The function block has two travel limits: son and soff. Either both, one, or neither of these limits can be configured for use with the sonu and soffu configuration parameters.

Automatic controls are connected to :a –input in the application. :m –input is for control in manual mode and it is very often left unconnected so that operator is able to control device from control room.

Control mode is chosen by :ma -input (0=manual, 1=automatic). If :ma –input remains unconnected, control mode can be done from control room application. If external signal is used, it is preferable to use pulse controls/triggers (ccob mode 1 or 2) instead of continuous writing to input so that operator is able to change control mode to manual when necessary.

ton Defines the response time to ON control. If the valve is opened successfully within the specified time, no other action is taken. If the OPEN position limit is not in use, parameter ton defines the time during which the valve is in the OPENING status. Then ton defines the operating time in the opening direction.

toff Defines the response time to OFF control. If the valve is closed successfully within the specified time, no other action is taken. If the CLOSED position limit is not in use, parameter toff defines the time during which the valve is in the CLOSING status. Then toff defines the operating time in the closing direction.

sonu Status ON (son) used. Set to 0 if ON-limit switch is not available

sofu Status OFF (soff) used. Set to 0 if OFF-limit switch is not available

ma Auto/Manual selection. 0=manual control mode, 1=automatic control mode

m control request in manual mode (1=open, 0=close)

a control request in automatic mode (1=open, 0=close)

Motor Control Function Block - mtr

pr:TAG_CODE		identifier	prompt	Value
ins :m :a :f :a1 :a2 :cur :e1 :e2 :e3 :e4 :e5 :e6 Customer	1 mtr	SORDER	EXECUTION ORDER	1
		SPORT	DIRECT ACCESS PORT NAME	pr:TAG_CODE
		TON=	ton TIME TO ON	4.0
		TOFF=	toff TIME TO OFF	4.0
		AFTFC=	aftfc AFTER FORCED CONTR	2
		CUU=	cuu CURRENT USED	1
		TTCUR=	ttcur TIME TO CURRENT AL	5.0
		CURHYST=	curhyst HYSTERESIS	1.0
		MAC<	mac MANUAL -> AUTO	1
		AMC<	amc AUTO -> MANUAL	1
		CURH<	curh HIGH LIMIT	100.0
		CURHH<	curhh HIGHER HIGH LIMIT	100.0

The motor control function block drives a binary output (outputs in pulse control mode) ON or OFF; the effect of the output on the controlled device is read back as feedback in the ON/ OFF format also. The feedback data gives the status of the controlled device, and it shows whether the control takes effect within the specified response time

Normal controls (in any mode) can be overridden with forced control **fon** and **foff**:

- fon = Forced ON
- foff = Forced OFF

In conflict situations foff dominates.

Inputs e1...e6 are available for fault, safety switch and other corresponding data. These inputs indicate that the controlled device has stopped due to an external cause, or that the device must be stopped. These inputs operate in the same way as foff , but overtake forced-control functions in order of priority

Automatic controls are connected to :a –input in the application. :m –input is for control in manual mode and it is very often left unconnected so that operator is able to control device from control room.

Control mode is chosen by :ma -input (0>manual, 1=automatic). If :ma –input remains unconnected, control mode can be chosen from control room application. If an external signal is used, it is preferable to use pulse/trigger (ccob mode 1 or 2) instead of continuous writing to input so that operator is able to change control mode to manual when necessary.

ton

Type: float
Default: 4.0
Description: Time to ON [s]

Defines the response time to ON control. If the motor is started successfully within the specified time, no other action is taken.

toff

Type: float
Default: 4.0
Description: Time to OFF [s]

Defines the response time to OFF control. If the motor is stopped successfully within the specified time, no other action is taken.

aftfc

Type: uns8
Default: 0
Description: Operation during and after forced control

Besides the forced control action, forced control in either direction (foff and fon) will affect the state of the ma input, depending on the value of the aftfc parameter.

The effect of the different values: 0 = motor is forced to the ma = MANUAL status for the duration of forced control

1 = motor is forced to the ma = MANUAL status in the execution cycle during which forced control came into effect

2 = motor is forced to the ma = MANUAL status in the execution cycle during which forced control came into effect, if the forced control affected the state of the motor

3 = forced control does not affect the state of the ma input

Others = error, which is interpreted as 0

aftfc does not affect an operation caused by external inputs e1...e6.

cuu

Type: bool8
Default: 0
Description: Current used

Defines whether motor current and the corresponding alarms are in use:

0 = motor current not used

1 = motor current and corresponding alarms are in use; the curh and curhh limits will be checked

mac

Type: bin
Default: 1
Description: Manual-to-auto change allowed

Permits switching from manual to automatic mode.

amc

Type: bin
Default: 1
Description: Auto-to-manual change allowed

Permits switching from automatic to manual mode.

More detailed manuals for all function blocks can be found on the remote desktop:

The diagram illustrates the process of finding detailed manuals for Valmet DNA function blocks. It starts with a Windows taskbar on the left, where the 'DNA Manuals' icon is highlighted with a red arrow. A green arrow points from this icon to a web browser window in the center. The browser window displays the 'Valmet DNA Manuals' index page, with a red arrow pointing to the 'PCS Function Blocks (ZIP)' link under the 'Application Libraries' section. Another green arrow points from this link to a third window on the right, which shows a detailed list of 'PCS Function Blocks' including various function block names like `ahs`, `ahs2`, `am`, `am2`, `apid`, `askfb`, `bhs`, `bhs2`, `bm2`, `btoicnv`, `coa`, `ccob`, `ccobe`, `ccobo`, `ccofa`, `ccofl`, `ccol`, `ccos`, `ccoa2`, `ccobe2`, `ccole2`, `ccose2`, `ccoti2`, `ccob2`, `ccoi8`, `ccoi16`, `ccoi32`, `ccou8`, `ccou16`, `ccou32`, `ccobl8`, `ccobl16`, `ccotx16`, `cfab`, `cnq`, `cnq2`, `cnt`, `comprs`, `cselt`, and `cwsqrt`.

Windows Taskbar (Left):

- UltraVNC
- Valmet DNA
- Loop Auto Tuner
- ACN Boot Media Configurator
- ACN Boot Media Configurator Doc...
- ACN IO Browser
- ACN IO Browser Help
- Backup Server Manager
- BU Installation document
- DiagLis
- DiagLis Users Guide
- DNA Engineering Settings
- DNA Explorer
- DNA Manuals** (highlighted with a red arrow)
- DNA Network Designer
- DNA Operate Client
- Dna System Configurator

Valmet DNA Manuals (Center):

- Expand All | Collapse All
- Operator Manuals
 - Operate
 - Report
 - Machine Monitoring
- Application Tools
- Application Libraries
 - Function Blocks
 - PCS Function Blocks (ZIP)** (highlighted with a red arrow)
 - DMM Function Blocks (PDF)
 - QCS Function Blocks
 - Prog2 Function Block
 - Fuzzy Function Blocks
 - PLU and MCP
 - Recipe Control
 - Type Collections
 - Field Bus Interfaces
 - Interfaces
- System Administration
- Hardware
- Declarations and Certificates

PCS Function Blocks (Right):

- Function Blocks
 - `ahs`
 - `ahs2`
 - `am`
 - `am2`
 - `apid`
 - `askfb`
 - `bhs`
 - `bhs2`
 - `bm2`
 - `btoicnv`
 - `coa`, `ccob`, `ccobe`, `ccobo`, `ccofa`, `ccofl`, `ccol`, `ccos`
 - `ccoa2`, `ccobe2`, `ccole2`, `ccose2`, `ccoti2`, `ccob2`
 - `ccoi8`, `ccoi16`, `ccoi32`, `ccou8`, `ccou16`, `ccou32`, `ccobl8`, `ccobl16`, `ccotx16`
 - `cfab`
 - `cnq`
 - `cnq2`
 - `cnt`
 - `comprs`
 - `cselt`
 - `cwsqrt`

You find DNA manuals here!